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Research Paper

Enhancing mesophilic methanogenesis in oleate-rich environments through optimized micro-aeration pretreatment

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ABSTRACT

Micro-aeration pretreatment has emerged as a promising technology for improving the performance of anaerobic bioreactors in the treatment of lipid-rich organic waste, particularly in mitigating the accumulation of long-chain fatty acids (LCFAs). Micro-aeration intensity is a critical factor in optimizing substrate hydrolysis and methanogenesis efficiency. In this study, optimal micro-aeration intensities for acetoclastic (30 mL-air/g-COD) and overall methanogenesis (7.5 mL-air/g-COD) were initially determined using acetate and glucose as substrates, respectively. Subsequently, the addition of 0.5 mM oleate (a typical LCFA) increased cumulative methane production by 22.1 % when acetate was used as the substrate after 30 mL-air/g-COD micro-aeration pretreatment. Conversely, it decreased cumulative methane production by 17.3 % when glucose was used as the substrate after 7.5 mL-air/g-COD micro-aeration pretreatment. Additionally, the population of facultative hydrolysis microorganisms, such as the genus *Pseudomonas*, increased by 25.7 % and 27.8 % when acetate and glucose were used as substrates, respectively. Furthermore, the predominant methane-producing archaea, including the genus *Methanosarcina*, increased by 27.3 % when acetate was used as the substrate, while the genus *Methanosaeta* decreased by 65.3 % when glucose was used as the substrate. Collectively, these findings provide insights into the methanogenesis pathway under optimal micro-aeration pretreatment conditions, guiding future research in this field.

1. Introduction

Anaerobic digestion (AD) is a promising technology for treating lipid-rich organic waste (Hu et al. 2022). Theoretically, 94.8 % of lipids can be converted into methane during the AD process, according to the McCarty equation (Jeganathan et al. 2006). This makes lipids a more potent methane-producing substrate (1010 mL CH_4/g VSS) compared to cellulose (379 mL CH_4/g VSS), protein (496 mL CH_4/g VSS) and glucose (351 mL CH_4/g VSS) (Raposo et al. 2012). During lipid hydrolysis, long-chain fatty acids (LCFAs) are produced. Nevertheless, LCFAs are recognized as common inhibitors and pollutants in AD process (Elsamadony et al. 2021). Previous studies have shown that LCFAs often accumulate in AD reactors due to the thermodynamically unfavorable and energy-requiring β-oxidation pathway, which required a prolonged duration (Saha et al. 2019). Additionally, LCFAs can adhere to microbial cell membranes, disrupting microbial homeostasis and hindering the vital movement of methane-producing archaea (MPA) (Amha et al. 2017, Kurade et al. 2019). To mitigate LCFA accumulation, researchers have explored strategies such as the supplementation of cations and natural adsorbents (Dereli et al. 2014, Nordell et al. 2013). However, the precise dosing of metal cations and the potential decomposition of insoluble compounds formed under thermophilic conditions can compromise the long-term stability of AD bioreactors (Soltan et al. 2019). Moreover, the dosage of natural adsorbents will augment accompanied by LCFAs enrichment, and the adsorbents completed by adsorption required subsequent treatment (Reeve and Fallowfield 2018). Therefore, it is worthwhile to investigate alternative approaches, such as enriching the abundance of functional microorganisms, to alleviate LCFA accumulation and enhance the overall efficiency of lipid-rich waste AD.

Previous studies have identified facultative anaerobic bacteria, such as genus *Pseudomonas* and *Clostridium* in phylum Firmicutes, as key

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microorganisms in mitigating LCFAs inhibition (Elsamadony et al. 2021). Recently, micro-aeration based AD system has been shown to effectively enrich genus *Pseudomonas* populations (Salome Duarte et al. 2018). In practice, this approach not only protects strict anaerobic microorganisms including MPA, but also maintains the stability and efficiency of lipid-based bio-methanation processes (Duarte et al. 2020, Salome Duarte et al. 2018). Furthermore, micro-aeration can enhance the performance of AD bioreactors in several aspects, including hydrogen sulfide removal, improved substrate hydrolysis efficiency and increased volatile fatty acids (VFAs) degradation (Nguyen and Khanal 2018). Nevertheless, comprehensive consideration of oxygen dosage and aeration strategy is crucial. Excessive oxygen can negatively impact the activity of MPA and disrupt their anaerobic environment. Conversely, insufficient oxygen dosage may have a negligible influence on bioreactor performance (Li et al. 2024, Wu et al. 2018). A previous study demonstrated that micro-aeration pretreatment, specifically the addition of 5 mL-oxygen/g-VS during the food waste digestion process, maximized cumulative methane production and increased the abundance of the phylum Firmicutes by 21 % compared to non-aerated conditions (Fu et al. 2016). Additionally, another research revealed that 15 mL oxygen dosage in a 1 L sequencing batch reactor resulted in a more than 10 % increase in cumulative methane production compared to non-aerated conditions, outperforming dosages of 5, 10 and 20 mL oxygen (Zhen et al. 2020). These findings highlight the critical role of micro-aeration intensity in pretreating organic waste for micro-aeration based AD systems. Nevertheless, optimal micro-aeration intensity can vary depending on the specific characteristics of the organic waste utilized as substrate and its impact on the methanogenesis process. To determine optimal micro-aeration intensities, typical substrates such as acetate and glucose can be employed. In contrast, when lipid-rich organic waste was used as a substrate, 24 mL-oxygen/g-VS micro-aeration intensity can exacerbate VFAs accumulation as well as hinder acetate production (Rafieenia et al. 2017). Nevertheless, further research is needed to investigate the effects of lipid-rich organic waste on methanogenesis under various micro-aeration intensities. Therefore, oleate-rich environment is chosen as a representative lipid-rich environment because oleate is abundant in lipid-rich wastewater (Sun et al. 2023).

In this research, a biochemical methane potential (BMP) test was initially conducted using sodium acetate and glucose as substrates to determine the optimal micro-aeration intensity for acetoclastic and overall methanogenesis. Subsequently, another BMP test was performed to investigate the impact of oleate addition on the optimal micro-aeration intensity and methanogenesis process. Finally, acetoclastic and overall methanogenesis pathways under the optimal micro-aeration intensity were elucidated.

2. Materials and methods

2.1. Inoculum and substrate

The seed sludge was sourced from a mesophilic (35 ± 1°C) continuous stirred tank reactor (CSTR) with a 3-L working volume, operating at an organic loading rate of 4 g/L_R d. It was acclimated to an artificial food waste mixture comprising fourteen kinds of food based on a previous research (Lee et al. 2008). This artificial food waste contained moderate amounts of pork, egg and fish, resulting in a lipid content of 10.45 % ± 0.22 %, which can enhance microbial community's tolerance to lipid-rich environments. The seed sludge exhibited the following characteristics: total solids (TS) of 18.04 ± 0.15 g/L, volatile solids (VS) of 12.18 ± 0.55 g/L, COD of 11.15 ± 0.35 g/L, and ammonia concentration of 2.41 ± 0.05 g/L.

Before the formal experiment, a buffered solution was prepared to establish a suitable external environment for the growth of anaerobic microorganisms. The solution contained the following elements (mg/L): $MgCl_2·6H_2O$, 400; $CaCl_2·2H_2O$, 150; NH_4Cl, 500; K_2HPO_4, 872; KH_2PO_4, 680; $NaHCO_3$, 5000; $FeCl_2·4H_2O$, 2; $CoCl_2·6H_2O$, 0.17; $ZnCl_2$, 0.07; H_3BO_3, 0.06; $MnCl_2·4H_2O$, 0.611; $NiCl_2·6H_2O$, 0.04; $CuCl_2·2H_2O$, 0.027; $Na_2MoO_4·2H_2O$, 0.025; EDTA, 5. In addition, 4 g-COD/L acetate and 8 g-COD/L glucose were used as substrates respectively to supply carbon source for MPA during acetoclastic and overall methanogenesis.

2.2. Experimental design

The experiment was conducted in serum vial bottles with a working volume of 50 mL and total volume of approximately110 mL. Initially, each bottle was inoculated with 10 mL of seed sludge, 10 mL of buffered solution, 5 mL of substrate and distilled water, followed by a two-minute N_2 purge to remove residual air. Afterwards, all bottles were compressed with butyl stoppers and then sealed with aluminum crimp caps. Finally, a small air volume was injected into each bottle to establish diverse micro-aeration intensities. After this procedure, the sealed bottles were incubated under mesophilic (35 ± 1°C) condition in a shaker incubator (BS-100B, Shanghai BenTing, Inc., China) at 125 rpm.

Micro-aeration intensity and air dosage were calculated based on the amount of saturated dissolved oxygen and Henry's Law. The saturated dissolved oxygen concentration can be calculated using the following formula (Zhang, 1999):

$DO = p/p_0*477.8/(T + 32.26)$.

Where p represents local barometric pressure, p_0 represents standard barometric pressure, T represents the temperature, which is 35°C in this case.

To maintain oxygen equilibrium in both the liquid phase and headspace, a critical air volume of 3 mL was determined based on Henry's Law. This volume ensures sufficient oxygen for saturation in the liquid phase (approximately 0.25 mL) and the headspace (approximately 0.35 mL). To investigate the effects of varying micro-aeration intensities, four different air volumes (3, 6, 12 and 24 mL) were employed. These volumes corresponded to micro-aeration intensities of 15, 30, 60 and 120 mL-air/g-COD for acetoclastic methanogenesis, and 7.5, 15, 30 and 60 mL-air/g-COD for overall methanogenesis, respectively. Additionally, a non-aerated control group was included. Three parallel samples were prepared for each micro-aeration condition. The detailed overview of the experimental conditions is presented in Table. 1.

As for the pivotal parameters in each serum vial bottle, the sludge VS content was controlled at 4 g/L. Furthermore, the oleate concentration was set at 0.5 mM, aligning with its previous reported IC_{50} value for traditional AD processes (Li et al. 2023, Silva et al. 2016).

2.3. Analytical methods

Biogas production was measured by a glass syringe, while biogas composition was analyzed via a gas chromatograph (SP-6890, Shandong Lunan, Inc., China). COD value was determined by COD digestion apparatus and a matching COD measuring instrument (HACH Loveland, CO, USA). TS and VS were measured according to standard methods (American Public Health Association, 2012). VFA samples were pretreated by centrifugation at 10000 rpm for 10 min and filtration through 0.45 μm pore size filters. VFA concentrations were then detected by a gas chromatography (GC-2010, Shimadzu, CO, Japan). Coenzyme F_{420} concentration was measured with a UV–vis spectrophotometer (P1, Shanghai MAPADA, Inc., China) based on the methods described in a previous research (Wang et al. 2020). Microbial community analysis involved 16S rRNA cloning and sequencing. The analysis procedure involved microorganism DNA extraction, polymerase chain reaction amplification, electrophoretic separation and fragment sequencing. The primers used for high-throughput sequencing were 515FmodF and 806RmodR. Detailed procedures for 16S rRNA followed those outlined in a previous research (Jing et al. 2013). Significant differences (P-values) between acetoclastic and overall methanogenesis were calculated using Excel 2010 (Microsoft Office, 2010).

Table 1
Detailed experimental conditions.

Number	Substrate	Micro-aeration intensity (mL-air/g-COD)	Oleate concentration (mM)	Number	Substrate	Micro-aeration intensity (mL-air/g-COD)	Oleate concentration (mM)
A1	acetate	0	0	AO1	acetate	0	0.5
A2	acetate	15	0	AO2	acetate	15	0.5
A3	acetate	30	0	AO3	acetate	30	0.5
A4	acetate	60	0	AO4	acetate	60	0.5
A5	acetate	120	0	AO5	acetate	120	0.5
G1	glucose	0	0	GO1	glucose	0	0.5
G2	glucose	7.5	0	GO2	glucose	7.5	0.5
G3	glucose	15	0	GO3	glucose	15	0.5
G4	glucose	30	0	GO4	glucose	30	0.5
G5	glucose	60	0	GO5	glucose	60	0.5

3. Results and discussion

3.1. Cumulative biogas and methane production

Fig. 1 (a) illustrates cumulative methane production over a 42-day period using acetate as the substrate under different micro-aeration intensities, excluding the addition of 0.5 mM oleate. The results indicated that both 30 and 15 mL-air/g-COD micro-aeration intensities promoted cumulative methane production by 5.2 % and 0.4 %, respectively, compared to non-aerated condition. Conversely, 120 mL-air/g-COD micro-aeration intensity hindered acetoclastic methanogenesis because excessive air addition contributed to more reactive oxygen species production exceeding microbial tolerance, which inhibited MPA methanogenic activity. These findings suggest that moderate micro-aeration intensities could enhance acetoclastic methanogenesis efficiency, with 30 mL-air/g-COD emerging as a potential optimal intensity.

On the other hand, as illustrated in Fig. 1 (b), while micro-aeration based AD system could theoretically enhance microbial tolerance, particularly for MPA, to oleate, the presence of oleate extended the lag phase of methanogenesis from 5 to 21 days, confirming its inhibitory

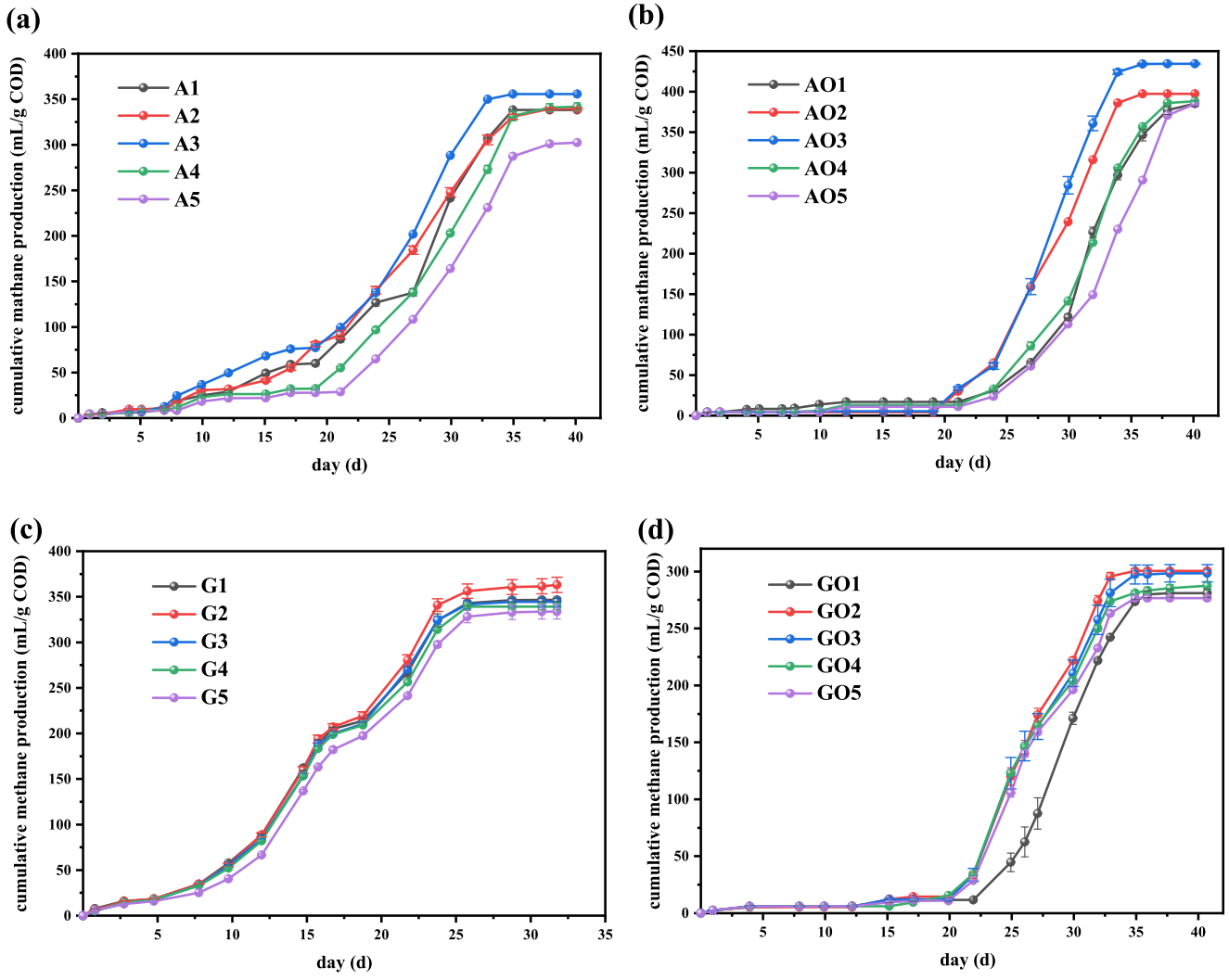


Fig. 1. Cumulative methane production (a) acetate as substrate without oleate addition (b) acetate as substrate with oleate addition (c) glucose as substrate without oleate addition (d) glucose as substrate with oleate addition.

characteristic during AD process (Cavaleiro et al. 2016, Salome Duarte et al. 2018). Notably, micro-aeration intensities of 30 and 15 mL-air/g-COD, in the presence of 0.5 mM oleate, resulted in a 12.8 % and 3.2 % increase in cumulative methane production, respectively, compared to non-aerated condition. Meanwhile, a comparative analysis of Fig. 1 (a) and (b) revealed that the addition of 0.5 mM oleate consistently promoted cumulative methane yield under identical aeration intensities. For instance, the addition of 0.5 mM oleate increased cumulative methane production from 355.76 ± 1.38 mL/g-COD to 388.28 ± 0.79 mL/g-COD after 30 mL-air/g-COD micro-aeration pretreatment. Further analysis indicated that the initial 4 g-COD/L acetate underwent complete methanogenesis because the cumulative methane production approached the theoretical methane production potential, with the additional methane production attributed to the biodegradation of 0.5 mM oleate. The COD value presented in Fig. 1 was based solely on the COD concentration of acetate, not the combined concentration of acetate and oleate. Collectively, these findings suggest that oleate can be decomposed by functional hydrolysis microorganisms to supply carbon source for methane synthesis, even in the absence of additional external carbon sources.

On the other hand, Fig. 1 (c) illustrates the cumulative methane production using glucose as the substrate without the addition of 0.5 mM oleate. Notably, the introduction of 0.5 mM oleate significantly extended the lag phase of methanogenic pathway from 1 to 21 days. In addition, both 7.5 and 15 mL-air/g-COD micro-aeration intensities enhanced cumulative methane production while 60 mL-air/g-COD micro-aeration intensity inhibited the methanogenic process compared to the non-aerated condition. In specific, 7.5 and 15 mL-air/g-COD micro-aeration pretreatments increased cumulative methane production by 4.8 % and 0.4 %, respectively. Based on these results, 7.5 mL-air/g-COD micro-aeration intensity can be identified as the optimal one for overall methanogenesis. Furthermore, in the presence of 0.5 mM oleate, 7.5 and 15 mL-air/g-COD micro-aeration intensities increased cumulative methane production by 3.5 % and 2.4 %, respectively. However, with the comparison of Fig. 1 (c) and (d), it was found that the addition of 0.5 mM oleate contributed to a 16.40 ± 1.90 % decrease in cumulative methane production compared to oleate-free condition. This conclusion contrasted with the observations made when using acetate as the substrate. This discrepancy can be attributed to the more complex methanogenic pathway of glucose compared to acetate. Added oxygen can be consumed to promote glucose hydrolysis and acidogenesis. In the presence of oleate, other functional microorganisms might consume the initial added oxygen to hydrolyze oleate, leading to oleate accumulation and subsequent inhibition of MPA activity. Therefore, when considering micro-aeration pretreatment in oleate-rich environments, the choice of substrate for MPA growth should be carefully considered due to their varying biodegradation pathways (Oh et al. 2020).

As presented in Table. 2, the specific methanogenic activity (SMA) results generally aligned with the cumulative methane production observed in all serum vial bottles generally. A significant difference ($P < 0.05$) in methanogenic activity was noted between acetoclastic and overall methanogenesis. The SMA values indicated that the activities of MPA were the highest among the three parallel samples in AO3 and G2 when acetate and glucose were used as substrates, respectively. In addition, when acetate was used as the substrate, the presence of oleate reduced methanogenesis activity under non-aerated conditions, while it enhanced methanogenic activity through moderate micro-aeration pretreatment. Conversely, when glucose was used as the substrate, the presence of oleate decreased overall methanogenic activity, regardless of aeration. This could be attributed to the relatively short hydrolysis duration of carbohydrates like glucose compared to lipids like oleate (Leung and Wang 2016). Therefore, during the glucose hydrolysis phase, oleate might not be sufficiently decomposed, leading to its accumulation around MPA and impeding its methanogenic activity. In addition, though the organic loading rate of the sludge used for glucose was twice that of acetate, it did not exceed the tolerance of the seed sludge, which could be stably cultivated at an organic loading rate of 4 g/L_R d. This accounted for the decline in cumulative methane production observed in the presence of oleate compared to the oleate-free condition under the same micro-aeration intensity when glucose was used as the substrate. Therefore, the optimal utilization of micro-aeration pretreatment must be comprehensively considered when selecting diverse complex organic compounds utilized as substrates for MPA growth and reproduction under oleate-rich conditions.

Table 2
Specific methanogenic activity (g-COD/g-VS) under each condition.

micro-aeration intensity (mL-air/g-COD)	acetate as substrate without oleate addition	acetate as substrate with oleate addition	micro-aeration intensity (mL-air/g-COD)	glucose as substrate without oleate addition	glucose as substrate with oleate addition
0	0.966 ± 0.007	1.100 ± 0.026	0	1.979 ± 0.026	1.563 ± 0.021
15	0.970 ± 0.017	1.135 ± 0.010	7.5	2.062 ± 0.116	1.717 ± 0.027
30	1.016 ± 0.019	1.241 ± 0.012	15	1.969 ± 0.072	1.699 ± 0.118
60	0.977 ± 0.026	1.109 ± 0.001	30	2.039 ± 0.036	1.606 ± 0.027
120	0.864 ± 0.065	1.099 ± 0.001	60	1.903 ± 0.111	1.580 ± 0.033

3.2. SCOD value and coenzyme F_{420}

Soluble chemical oxygen demand (SCOD) values represent the concentration of dissolved organic compounds, which can serve as carbon sources for the growth of MPA. Generally, MPA exhibited the highest activity under the peak value of SCOD and the duration required to reach this peak value represented the lag phase of the methanogenesis (Zhen et al. 2020). As presented in Fig. 2, SCOD value exhibited a continuous downward tendency in the bottles without oleate addition. In contrast, samples with oleate addition showed a slight initial increase followed by a subsequent decrease. This suggested that organic compounds, such as acetate and glucose, could be continuously utilized by relevant functional anaerobic microorganisms in the absence of LCFAs. On the other hand, LCFAs like oleate could be decomposed by facultative hydrolytic anaerobic microorganisms during the lag phase (day 1 to 21), providing additional SCOD. Micro-aeration pretreatment could further enhance oleate hydrolysis, contributing to the slight increase in SCOD values. Subsequently, bioavailable organic compounds were continuously utilized by anaerobic microorganisms from day 21 to 42, resulting in a consecutive decrease in SCOD values. Therefore, the overall trends of SCOD values were consistent with the cumulative methane production as depicted in Fig. 1. This indicated that micro-aeration based AD pretreatment could promote the biodegradation of organic compounds under moderate micro-aeration intensities.

Though cumulative methane production stabilized after 42 days of cultivation, the amount of bioavailable organic compounds remaining in the digestion slurry after the experiment warranted further investigation. Analysis of VFA types revealed that only a small amount of acetate and no other types of VFA. The concentration of acetate in each serum vial bottle remained below 20 mg/L, regardless of the addition of oleate. In specific, the concentration of acetate under each experimental condition ranged from 8.8 to 19.7 mg/L (equivalent to 8.2 to 18.5 mg-COD/L), significantly lower than the initial supplied SCOD. Therefore, it can be anticipated that the supplied carbon sources, including glucose and acetate, had been fully utilized by functional anaerobic microorganisms. In addition, the absence of VFA types other than acetate in the bottles using acetate as the substrate indicated that oleate was decomposed into smaller, bioavailable organic compounds. Other fermentative microorganisms utilized these compounds to produce acetate or hydrogen, which subsequently served as substrates for methane production.

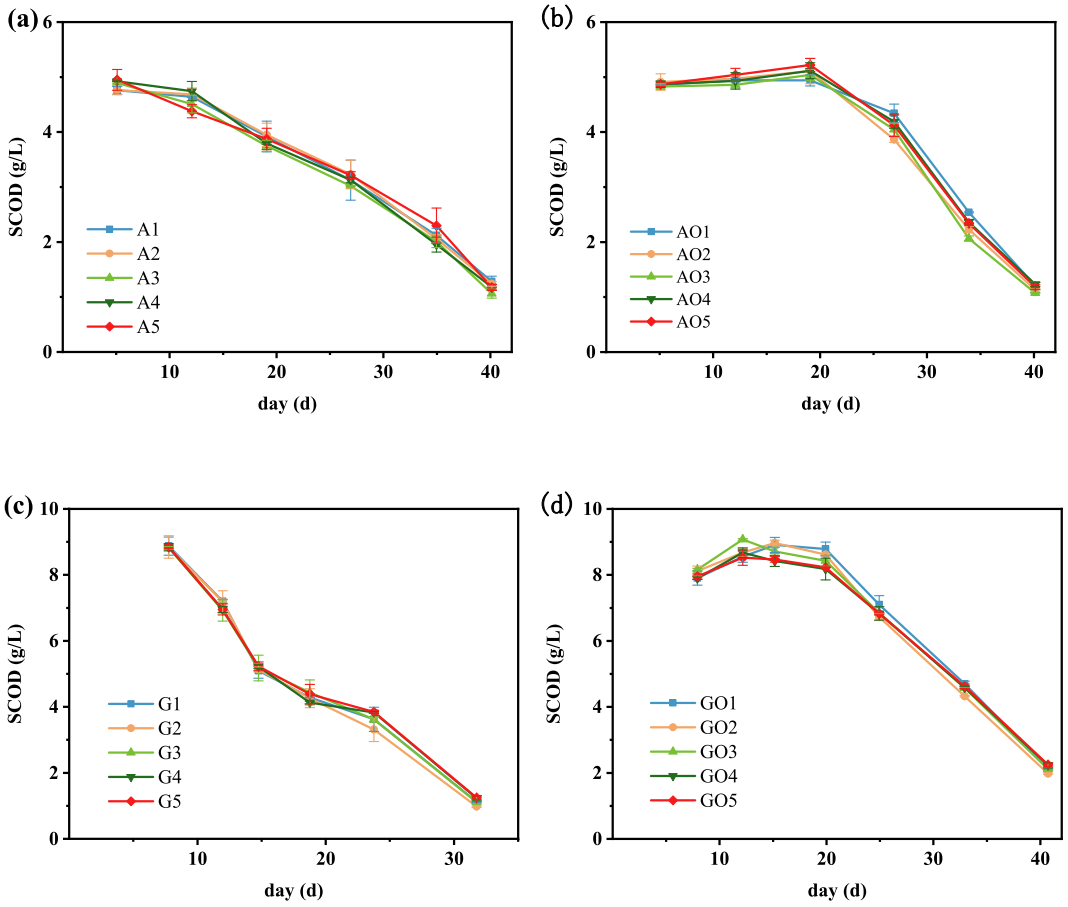


Fig. 2. SCOD value under different conditions (a) acetate as substrate without oleate addition (b) acetate as substrate with oleate addition (c) glucose as substrate without oleate addition (d) glucose as substrate with oleate addition.

As a typical electron carrier involved in both anabolic and catabolic redox reactions within the hydrogenase system, coenzyme F_{420} can play a crucial role in the growth and reproduction of hydrogen-utilizing MPA (Yu et al. 2023). Therefore, the concentration of coenzyme F_{420} could serve as a proxy for MPA activity (Zhang et al. 2022). In general, the trends in coenzyme F_{420} concentration observed in each serum vial bottle aligned with the trends in cumulative methane production. According to the analysis of Fig. 3, when acetate was used as the substrate, the bottles treated with 30 mL-air/g-COD micro-aeration pretreatment exhibited the highest coenzyme F_{420} concentration (0.516 μmol/L), while when glucose was used as the substrate, the bottles treated with 7.5 mL-air/g-COD micro-aeration intensity showed the highest coenzyme F_{420} concentration (0.635 μmol/L). This could be attributed to the fact that excessive oxygen addition, such as 60 mL-air/g-COD micro-aeration, could inhibit MPA activity and consequently suppress coenzyme F_{420} synthesis. Conversely, inadequate air addition, such as 15 mL-

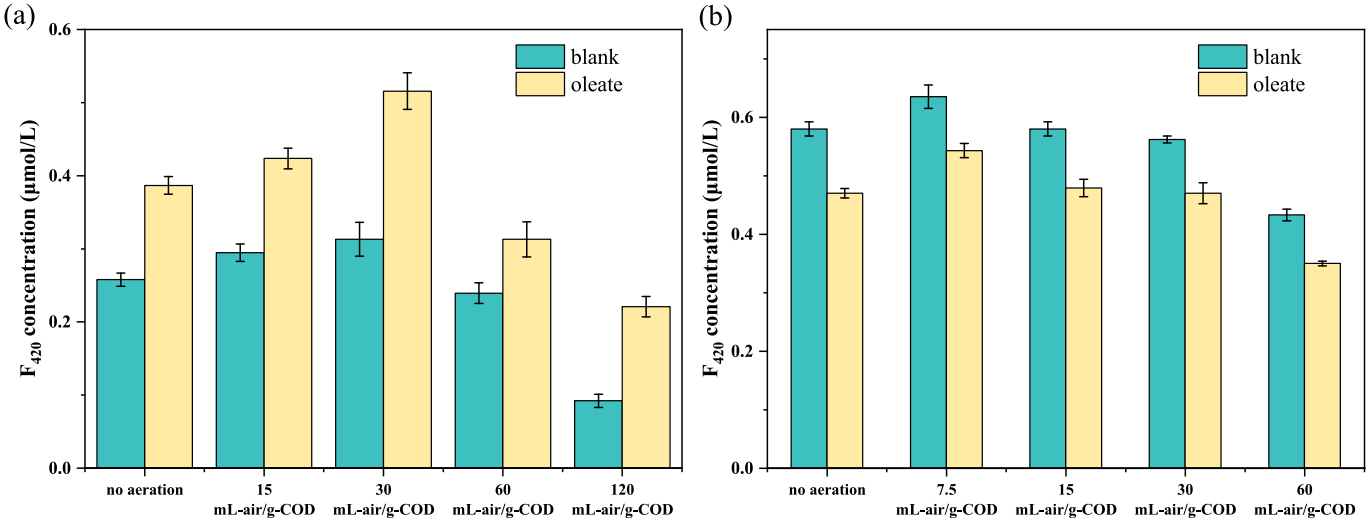


Fig. 3. Coenzyme F_{420} concentration after the experiment (a) acetate as substrate (b) glucose as substrate.

air/g-COD micro-aeration, might not significantly promote coenzyme F_{420} synthesis for acetoclastic methanogenesis. Furthermore, the addition of 0.5 mM oleate could hinder the synthesis of coenzyme F_{420} for overall methanogenesis, which can be ascribed to its incomplete degradation and excessive accumulation. However, it could promote the synthesis of coenzyme F_{420} for acetoclastic methanogenesis, which might be attributed to the additional carbon source provided by oleate degradation. Therefore, it was inferred that 30 mL-air/g-COD micro-aeration intensity was optimal for acetoclastic methanogenesis, while 7.5 mL-air/g-COD was optimal for overall methanogenesis.

3.3. Microbial composition

According to the 16S rRNA gene sequencing results, the bacterial community in the seed sludge at the phylum level was primarily composed of Proteobacteria (29.7 %), Bacteroidota (21.2 %), Actinobacteriota (12.5 %), Thermotogota (10.7 %) and Firmicute (9.3 %), as shown in Fig. 4 (a). These phyla are commonly found in anaerobic bioreactors utilized for food waste treatment (Afrizal et al. 2022, Xu et al. 2022). After more than 35 days cultivation, the total percentage of Firmicute and Thermotogota achieved a significant increase from 20.0 % to over 35.8 %, indicating that several genera within these phyla could survive in the provided environment, both in the presence and

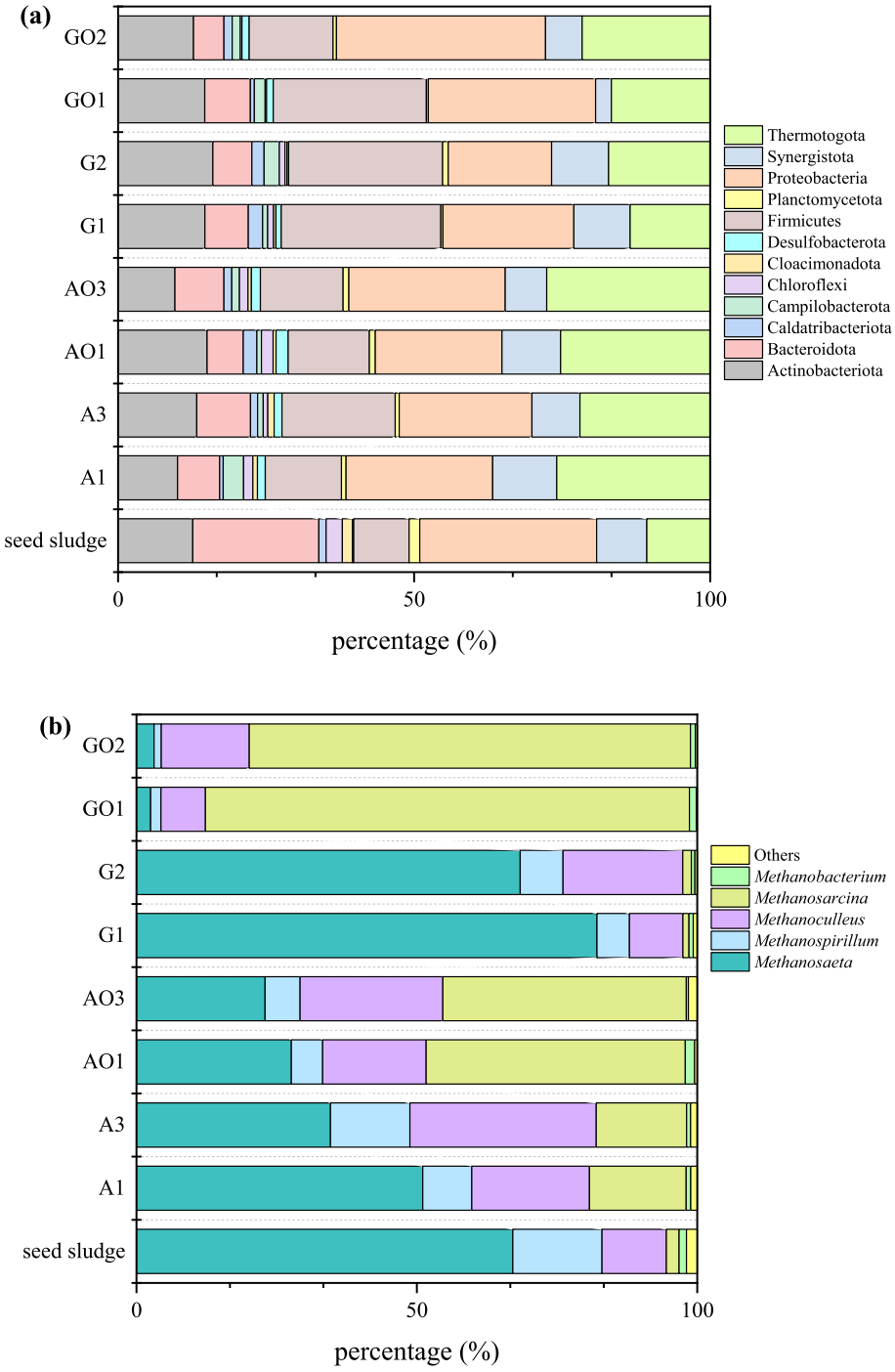


Fig. 4. Relative abundance of microorganism in seed and post-experiment sludge (a) bacteria composition at the phylum level (b) archaea composition at the genus level (c) microbial composition at the genus level.

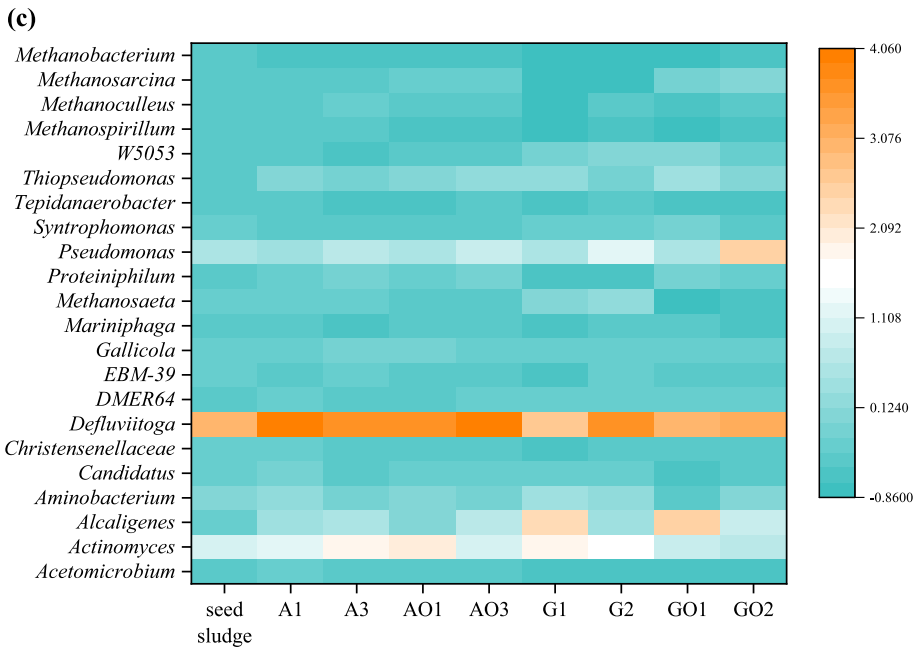


Fig. 4. (*continued*).

absence of oleate. Nevertheless, the proportion of phylum Bacteroidota decreased from over 21.1 % in the seed sludge to a range of 5.1 % to 9.1 % after the experiment. No significant changes were observed in the proportion of other phyla. According to previous studies, an increase in the abundance of Firmicute and Thermotogota had been observed in food waste AD bioreactor after micro-aeration pretreatment, and these phyla can play the role in hydrolysis and fermentation processes through their functional enzymes (Salome Duarte et al. 2018, Wang et al. 2022). Furthermore, the incremental percentage of Firmicute in the samples utilizing glucose as the substrate was higher than that in the samples utilizing acetate as the substrate, which aligned with their role in glucose hydrolysis process. It can be attributed to that glucose should undergo hydrolysis to produce acetate or carbon dioxide/hydrogen for subsequent methanogenesis.

On the other hand, the predominant archaea in the seed sludge at the genus level principally consisted of *Methanosaeta* (68.4 %), *Methanospirillum* (16.2 %), and *Methanoculleus* (11.7 %), as depicted in Fig. 4 (b). As the most abundant archaeon in the seed sludge, *Methanosaeta* could utilize acetate as carbon source for methane synthesis via the acetoclastic methanogenesis pathway (Wang et al. 2024). Other predominant MPA, including *Methanospirillum* and *Methanoculleus*, could produce methane through hydrogenotrophic methanogenesis pathway (Jain et al. 2021, Ni et al. 2022). Nevertheless, after the experiment, a significant increase in the abundance of the predominant MPA (genus *Methanosarcina*) was observed, especially in the bottles with oleate addition. This suggested that genus *Methanosarcina* could outcompete other acetate-utilizing MPA, such as *Methanosaeta*, under oleate-rich conditions. Moreover, the percentages of acetate-utilizing MPA including *Methanosaeta* and *Methanosarcina* and hydrogen-utilizing MPA including *Methanospirillum* and *Methanoculleus* remained relatively stable in the samples utilizing acetate as the substrate. Nevertheless, when glucose was used as the substrate, the percentage of hydrogen-utilizing MPA decreased while the percentage of acetate-utilizing MPA increased, indicating that acetate was produced in greater quantities than hydrogen/carbon dioxide during glucose hydrolysis.

At the genus level, genus *Defluviitoga* was the most abundant genus in all serum vial bottles. During glucose hydrolysis, *Defluviitoga* has been proved to play the role in producing small molecular organic compounds such as acetate, carbon dioxide and hydrogen, which serve as substrates for MPA (Li et al. 2019). In addition, under the optimal micro-aeration intensity, there was a significant increase in the abundance of genus *Pseudomonas*, a genus known to survive in both anaerobic and aerobic environments (Pedraz et al. 2020). This phenomenon was consistent with previous studies that identified genus *Pseudomonas* as the functional anaerobic microorganism involved in mitigating LCFA accumulation in micro-aeration based AD environments (Salome Duarte et al. 2018). Other abundant genera, including *Actinomyces* and *Alcaligenes*, also exhibited increased abundance after the experiment. These genera have been previously characterized as facultative anaerobic microorganisms (Corcione et al. 2018, Wang et al. 2021). Therefore, in oleate-rich environments with optimal micro-aeration pretreatment, certain genera, such as *Pseudomonas*, could contribute to alleviating oleate inhibition and ensuring the continuity of the methanogenesis pathway.

3.4. Methanogenesis pathway

Based on the aforementioned analysis, Fig. 5 illustrates the possible methanogenesis process under oleate-rich conditions with optimal micro-aeration pretreatment. In the bottles utilizing acetate as the substrate with 30 mL-air/g-COD micro-aeration pretreatment, oleate could be decomposed by functional hydrolytic microorganisms such as those belonging to the genus *Pseudomonas*. The optimal micro-aeration pretreatment could increase the abundance of genus *Pseudomonas* by 23.7 %, thereby promoting oleate hydrolysis process. The by-products generated from oleate hydrolysis could be converted into acetate through acidogenesis, providing the substrate for acetate-utilizing MPA including *Methanosaeta* and *Methanosarcina*. Consequently, the abundance of these microorganisms increases, contributing to enhanced acetoclastic methanogenesis and a 22.1 % increase in cumulative methane production.

On the other hand, in the bottles utilizing glucose as the substrate with 7.5 mL-air/g-COD pretreatment, oleate could be hydrolyzed by genera such as *Pseudomonas*, while glucose could be biodegraded by genera such as *Defluviitoga*. In specific, micro-aeration pretreatment can enrich the abundance of genera *Pseudomonas* and *Defluviitoga* by 27.8 % and 6.3 %, respectively. Nevertheless, the hydrolysis duration of glucose was shorter than that of oleate. This contributed to the accumulation of oleate around MPA before their hydrolysis. Furthermore, a higher proportion of MPA transitioned to *Methanosarcina* rather than *Methanosaeta*. Therefore, this hindered MPA methanogenic activity and

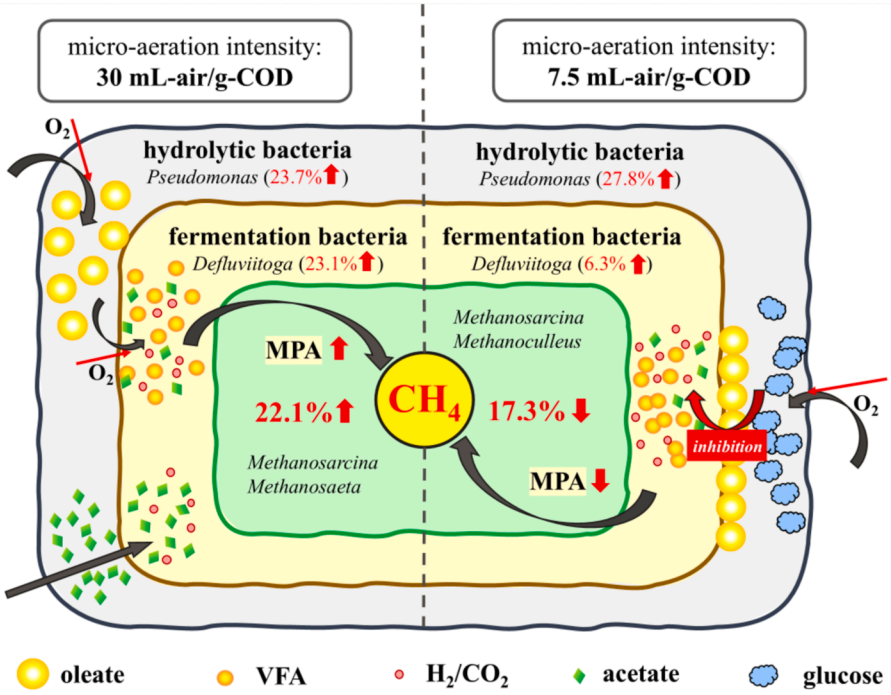


Fig. 5. Methanogenesis pathway with the addition of diverse substrates (acetate and glucose) respectively under the optimal micro-aeration intensity (The red arrow represents the percentage compared with the sample without oleate addition).

resulted in a 17.3 % decrease in cumulative methane production.

4. Conclusion

Micro-aeration based AD technology is a promising approach to mitigate LCFA accumulation and regulate biogas production, particularly for pretreating lipid-rich organic waste. Micro-aeration intensity can be recognized as a crucial parameter in this process. This research demonstrated that 30 mL-air/g-COD aeration was the optimal micro-aeration intensity for acetoclastic methanogenesis, while 7.5 mL-air/g-COD aeration was optimal for overall methanogenesis in the absence of oleate. On the other hand, the addition of 0.5 mM oleate promoted acetoclastic methanogenesis by 22.1 % but inhibited overall methanogenesis by 17.3 % under optimal micro-aeration conditions. Furthermore, this research revealed that moderate micro-aeration intensity accelerated the proliferation of facultative anaerobic bacteria, such as those belonging to the genera *Pseudomonas*, *Actinomyces* and *Alcaligenes*. Concurrently, the abundance of predominant MPA, including genus *Methanosarcina*, increased by 27.3 % during acetoclastic methanogenesis, while the abundance of genus *Methanosaeta* decreased by 65.3 % during overall methanogenesis. Therefore, appropriate micro-aeration intensity should be taken into systematical consideration to optimize methanogenesis or co-digestion efficiency. Overall, this study provides valuable insights into the optimal micro-aeration conditions for future research in this field.

CRediT authorship contribution statement

Xiang Li: Writing – original draft, Investigation, Formal analysis. **Yi-Juan Yan:** Visualization, Investigation, Data curation. **Hong-Ming Wu:** Software, Formal analysis, Data curation. **Samir Ibrahim Gadow:** Writing – review & editing, Validation, Supervision. **Hongyu Jiang:** Software, Investigation, Conceptualization. **Zhe Kong:** Writing – review & editing, Supervision. **Yong Hu:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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